Year 3 Mathematics Solutions

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



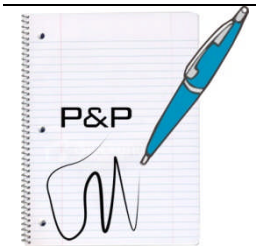
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



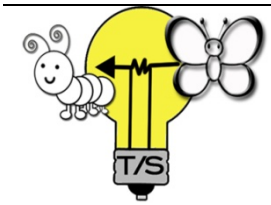
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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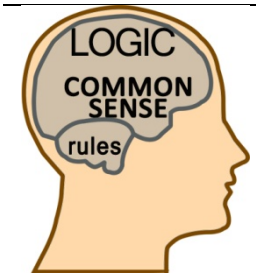
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

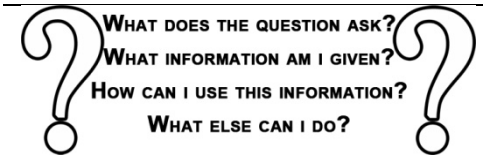
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 3 Mathematics

Solutions

Number

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Exercise 1

Representing Numbers

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1) Write as numbers

a) Twenty seven
27

b) Forty two
42

c) Ninety three
93

d) Twelve
12

e) Fifty
50

2) Write as numbers

a) One hundred and three
103

b) Two hundred and ninety seven
297

c) Six hundred and thirty three
633

d) Nine hundred and eleven
911

e) Three hundred and twenty
320

3) Write in words

a) 703
Seven hundred and three

b) 297
Two hundred and ninety seven

c) 333
Three hundred and thirty three

d) 90
Ninety

e) 201
Two hundred and one

f) 111
One hundred and eleven

g) 0
Zero

4) Write down the number that comes before each of these numbers

a) 33
32

b) 56
55

c) 105
104

d) 12
11

e) 171
170

f) 109
108

g) 243
242

h) 190
189

i) 900
899

j) 30
29

k) 1
0

l) 1000
999

5) Write the number that comes after each of these numbers

a) 19

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20

b) 109
110

c) 888
889

d) 223
224

e) 801
802

f) 711
712

g) 999
1000

h) 309
310

6) Put these numbers in order from smallest to largest

325, 101, 123, 1000, 946, 121, 15, 221, 323, 104, 694

15, 101, 104, 121, 123, 221, 323, 325, 694, 946, 1000

7) Put these numbers in order from largest to smallest.

201, 204, 402, 912, 911, 333, 322, 921, 221, 121, 4

921, 912, 911, 402, 333, 322, 221, 204, 201, 121, 4

8) What is the value of the number 4 in each of these numbers?

a) 104

4 ones

b) 435

4 hundreds

c) 214
4 ones

d) 427
4 hundreds

e) 4
4 ones

f) 40
4 tens

g) 204
4 ones

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Exercise 2

Addition & Subtraction

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1) Add these numbers

a) 32 + 14
46

b) 47 + 19
66

c) 62 + 35
97

d) 77 + 22
99

e) 13 + 17
30

f) 41 + 44
85

2) Add these numbers

a) 225 + 52
277

b) 432 + 41
473

c) 809 + 77
886

d) 435 + 23
458

e) 822 + 11
833

f) 934 + 73
1007

3) Subtract these numbers

a) 86 - 42
44

b) 54 - 42
12

c) 75 -51
24

d) 99 - 33
66

e) 54 - 12
42

f) 65 – 21
44

4) Peter has 40 cents, John has 25 cents. How much money do they have between them?

40 cents plus 25 cents equals 65 cents

5) Alan weighs 45 kg, Chris weighs 48 kg. How much do they weigh together?

45 kg plus 48 kg equals 93 kg

6) There are 15 more students in year 3 than in year 4. If there are 46 students in year 3, how many students are in year 4?

46 students plus 15 students equals 61 students

7) Tom and Jerry have read 40 books between them. If Tom has read 18 books, how many books has Jerry read?

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40 books subtract 18 books equals 22 books

8) 38 students passed a test, 12 failed, and 5 were absent. How many students are in the class?

38 students plus 12 students plus 5 students equals 55 students

9) What number is 43 less than 175?

175 subtract 43 equals 132

10) What is the difference between 210 and 344?

344 subtract 210 equals 134

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Exercise 3

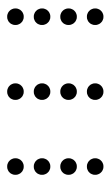
Multiplication & Division

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1) In each of the pictures below

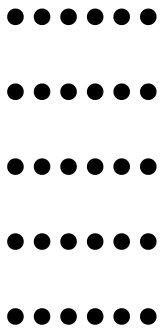
- How many dots in each row?
- How many rows are there?
- How many dots are there in total?

a)



4 in each row
3 rows
12 dots

b)



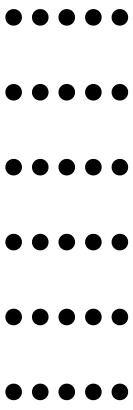
6 in each row
5 rows
30 dots

c)



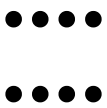
2 in each row
4 rows
8 dots

d)



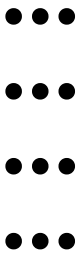
5 in each row
6 rows
30 dots

e)



4 in each row
2 rows
8 dots

f)



3 in each row
4 rows
12 dots

2) In question 1, which answers are the same? Why are they the same?

Answers a and f are the same

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Answers b and d are the same
Answers c and e are the same

They are the same because it doesn't matter which way round you multiply the numbers as long as they are the same numbers

3)

a) How many stars are there in the diagram?

******

******

******

18

b) How many lots of 6 are there?

3 lots of 6

c) How many lots of 6 in 18?

3

d) What is 18 ÷ 6?

3

e) How many lots of 3 are there?

6 lots of 3

f) How many lots of 3 in 18?

6

g) What is 18 ÷ 3?

6

4)

a) How many stars are there in the diagram?

****

****

****

****

****

20 stars

b) How many lots of 4 are there?

5 lots of 4

c) How many lots of 4 in 20?

5

d) What is 20 ÷ 4?

5

e) How many lots of 5 are there?

4

f) How many lots of 5 in 20?

4

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g) What is $20 \div 5$

4

5) Use the first 4 questions or any other way you know to answer these questions

a) 3×5

15

b) 5×3

15

c) $15 \div 5$

3

d) $15 \div 3$

5

e) $24 \div 6$

4

f) $24 \div 4$

6

g) 7×3

21

h) 4×8

32

i) $21 \div 7$

3

j) $32 \div 8$

4

6) Multiply the following

a) 9×5

45

b) 5×9

45

c) 8×4

32

d) 4×8

32

e) 7×6

42

f) 6×7

42

g) 3×15 (think of an easier way to do this)

Is the same as 15 x 3 = 45

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Or 3 x 15 = 3 x 3 x5 = 9 x 5 = 45

7) Mary has 4 lollies; Julie has 5 times as many. How many lollies does Julie have?

4 x 5 = 20 lollies

8) Alan wants to share his lollies amongst himself and his friends so everyone gets the same amount. He has 3 friends and 24 lollies. How many lollies does each person get?

24 ÷ 3 = 8

9) Kathy is having a birthday party and her mum wants to make sure there are enough cup cakes for everyone. She thinks each person will eat 3 cup cakes. If there are going to be a total of 11 people at the party how many cup cakes should Karen's mum make?

11 x 3 = 33

10) Every child in Tim's class received 4 pencils. If 32 pencils were given out, how many children in Tim's class?

32 ÷ 4 = 8 children

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Exercise 4

Number Patterns

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1) Fill in the blanks

a) 3, 6, 9, ___, 15

The numbers go up by 3, so missing number is 12

b) 2, 4, ___, 8, 10, ___

The numbers go up by 2, so the missing numbers are 6 and 12

c) 6, 12, 18, ___, ___

The numbers go up by 6, so the missing numbers are 24 and 30

d) ___, 14, 21, 28, 35, ___

The numbers go up by 7, so the missing numbers are 7 and 42

e) 4, 8, ___, ___, ___, 24

The numbers go up by 4, so the missing numbers are 12, 16, and 20

f) ___, 18, ___, 36, 45

The numbers go up by 9, so the missing numbers are 9 and 27

2) Fill in the blanks

a) 25, 20, ___, ___, 5, ___

The numbers go down by 5, so the missing numbers are 15, 10, and 0

b) 40, 32, ___, ___, 8

The numbers go down by 8, so the missing numbers are 24 and 16

c) 63, 54, 45, ___, ___, 18

The numbers go down by 9, so the missing numbers are 36 and 27

d) 63, 60, 57, 54, 51, ___, 45, ___, 39, ___, ___

The numbers go down by 3, so the missing numbers are 48, 42, 36, and 33

e) 14, 11, 8, ___, ___

The numbers go down by3, so the missing numbers are 5 and 2

3) Fill in the missing numbers

a) $2 \times 6 = 4 \times __$

2 x 6 = 12
4 x **3** = 12

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b) $5 \times 4 = 2 \times __$

5 x 4 = 20
2 x **10** = 20

c) $6 \times 6 = 9 \times __$

6 x 6 = 36
9 x **4** = 36

d) $4 \times 4 = 8 \times __$

4 x 4 = 16
8 x **2** = 16

e) $7 \times 6 = 6 \times __$

7 x 6 = 42
6 x **7** = 42

f) $3 \times 13 = 13 \times __$

3 x 13 = 39
13 x **3** = 39

4) For the given number, list all the numbers that divide into it
Example: 20
1, 2, 4, 5, 10, 20

a) 6

1, 2, 3, 6

b) 12

1, 2, 3, 4, 6, 12

c) 16

1, 4, 16

d) 20

1, 2, 4, 5, 10, 20

e) 25

1, 5, 25

f) 7

1, 7

g) 11

1, 11

h) What is special about the last two numbers?

Only one and itself divide into the number

5) Jane wants to share her lollies by giving 6 people 5 lollies each. One of the people doesn't want any. How can Jane share her lollies so everyone else gets the same amount?

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Jane must have 6 x 5 = 30 lollies. To share them among 5 people, each person must get 6 lollies (30 ÷5 = 6)

6) Tom walked 4 km per day for 6 days in a row. If Alan walks for 8 days, how many km per day should he walk to go the same total distance that Tom did?

Tom walked 4 x 6 = 24 km
Alan walks for 24 ÷ 8 = 3 km per day

7) Peter notices a pattern of fish in a row of fish tanks at the pet store. The first tank had 3 large fish in it. The second tank had 6 medium sized fish. The next tank had 9 smaller fish. There were 7 tanks in the row and the pattern continued to the last one.

a) How many fish were in the last tank?

The pattern is 3, 6, 9, ___, ___, ___, ___

The missing numbers are 12, 15, 18, and 21

b) How many fish in the whole row?

If you add the numbers together you get 84 fish

8) Graham makes a puzzle for his friends. In a crate he places 84 buttons; in the next one he places 77, then 70 in the next. If he continues this pattern:

a) How many buttons will be in the next crate?

The number of buttons decreases by 7 in each crate; so in the next crate there will be 63 buttons

b) How many buttons will be in the last crate?

The pattern continues until the last crate has 7 buttons

84, 77, 70, 63, 56, 49, 42, 35, 28, 21, 14, 7

c) How many crates will he use?
There are 12 crates with buttons in them

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Exercise 5

Fractions

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1) Write the following as a fraction

a) One half

$\frac{1}{2}$

b) One quarter

$\frac{1}{4}$

c) One eighth

$\frac{1}{8}$

d) Three quarters

$\frac{3}{4}$

e) Five eighths

$\frac{5}{8}$

f) Two quarters

$\frac{2}{4}$

2) Write the following in words

a) $\frac{1}{4}$

One quarter

b) $\frac{1}{2}$

One half

c) $\frac{3}{4}$

Three quarters

d) $\frac{1}{8}$

One eighth

e) $\frac{3}{8}$

Three eighths

3) Put these fractions in order from smallest to largest

$\frac{3}{4}, \frac{2}{4}, \frac{4}{4}, \frac{1}{4}$

$\frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{4}{4}$

4) Put these fractions in order from largest to smallest

$\frac{5}{8}, \frac{1}{8}, \frac{7}{8}, \frac{2}{8}, \frac{6}{8}$

$\frac{7}{8}, \frac{6}{8}, \frac{5}{8}, \frac{2}{8}, \frac{1}{8}$

5) Fill in the missing numbers

$\frac{1}{2}, \frac{2}{2}, \frac{3}{2}, \frac{4}{2}, __, __$

$\frac{5}{2}, \frac{6}{2}$

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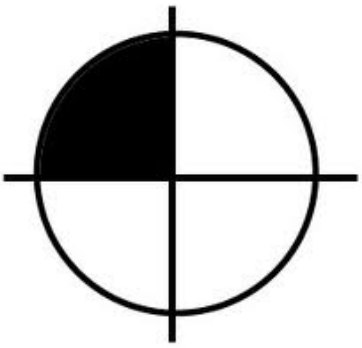
6) Fill in the missing numbers

$\frac{1}{8}, \frac{2}{8}, __, \frac{4}{8}, __, __$

$\frac{3}{8}, \frac{5}{8}, \frac{6}{8}$

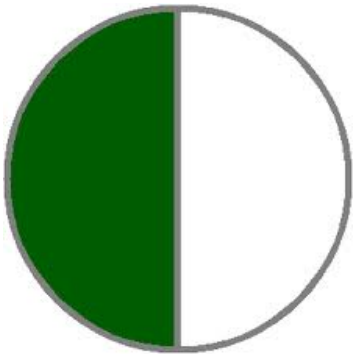
7) What fraction is shaded in the following diagrams?

a)



One quarter, $\frac{1}{4}$

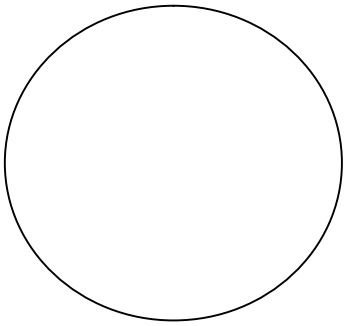
b)



One half, ½

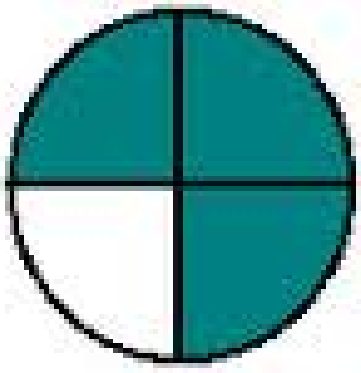
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c)



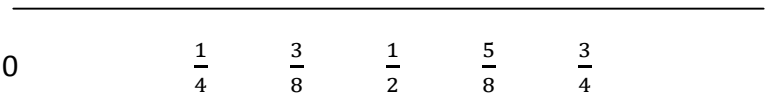
One whole (1)

d)



Three quarters, $\frac{3}{4}$

8) Place the fractions $\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{5}{8}, \frac{3}{8}$ on the number line



9) Tim has one quarter of his lollies left, while Jack has eaten three quarters. Who has more lollies left?

They have the same amount left. If Jack has eaten three quarters, he also has one quarter left

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Exercise 6

Chance

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1) Alan tosses a coin. What might the coin show when it lands?

Heads or tails

2) Peter rolls a dice. List all the numbers that he could get

1, 2, 3, 4, 5, 6

3) John has one of every coin in a bag. If he picks one without looking, list what coin he might pull out?

5 cents, 10 cents, 20 cents, 50 cents, 1 dollar, 2 dollars

4) Veronica has 9 tiles in a bag. Each tile has a different counting number written on it. List what tile she might pull out of the bag

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

5) There are 6 red shirts, 1 blue shirt and 15 yellow shirts in a draw. If a boy pulls a shirt out without looking:

a) List what colour shirt he might pull out

Red, blue or yellow

b) Which colour shirt will he probably pull out?

Yellow

c) Which colour shirt will he probably NOT pull out?

Blue

6) There are 20 red, 5 blue and 1 green lollies in a jar. If Jack closes his eyes and chooses one:

a) What colour lolly will he probably choose?

Red

b) What colour lolly would he be lucky to get?

Green

c) Is he more likely to get a green lolly or a yellow lolly?

Green, since there are no yellow lollies

d) Name a lolly colour that it would be impossible to get

Yellow or any colour that is not in the bag

7) In a jar there are 20 blue buttons. In another jar there are 10 blue and 10 yellow buttons.

a) From which jar would Colin be certain of picking a blue button with his eyes closed?

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The first jar has only blue

b) From which jar would be maybe get a yellow button?

The second jar has some yellow

c) From which jar would he definitely NOT get a yellow button?

The first jar has no yellow

d) Has he got more chance of picking a blue or yellow button from the second jar?

The same chance since there are equal numbers of them

8) Of the following events, which are certain to happen, impossible, or could happen?

a) The sun will rise tomorrow

Certain

b) You will eat food

Certain

c) You will go to school

Could happen (not on weekends or holidays)

d) You will get every maths question right

Could happen, but not very likely

e) You will turn 45 years old tomorrow

Impossible

f) Everyone in your class will win a million dollars tomorrow

Impossible

g) You will ride a bicycle

Could happen

9) Tom rolls a normal 6 sided dice. Which number is he most likely to roll?

Any of the numbers from 1 to 6 could come up

10) Alan tosses a coin; is it more likely to land on a head or a tail?

The coin is equally likely to land on a head or a tail.

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11) Peter spins a spinner with 3 red and 3 white faces. Which colour is he more likely to spin?

A red or white is equally likely since there are an equal number of faces

12) Peter spins a spinner with 1 red and 5 white faces. Which colour is he most likely to spin?

There are more white faces; therefore he has more chance of spinning white

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Year 3 Mathematics

Solutions

Data

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Exercise 1

Data Tables

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1) Tom made a table that shows how many of his classmates have each colour as their favourite

Red	Green	Yellow	Blue	White	Black
3	4	1	4	5	2

a) How many children in Tom's class?

19

b) Which colour was most popular?

White

c) Which colour was least popular?

Yellow

d) Which colours had equal numbers of children voting for it?

Blue and green

e) If one child had picked blue instead of white, would that change your answer to part b?

Yes, since the votes would be 5 for blue and 4 for white

2) A group of people was asked to vote for one day as their favourite day of the week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
1	3	5	10	5	6	15

a) How many people were asked?

45

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b) What was most people's favourite day?

Sunday

c) Why might this be?

Many people have this day off work and school

d) Which day do most people not like?

Monday (has least number of votes)

3) A man made a list of the cost of a type of blanket at different times of the year

January	March	May	July	September	November
$3.50	$4	$5	$6.50	$5	$4

a) In which of the months was the blanket the cheapest?

January

b) In which month was the blanket dearest?

July

c) What was the difference in its price between these 2 months?

$3

d) Name two months where the price was the same

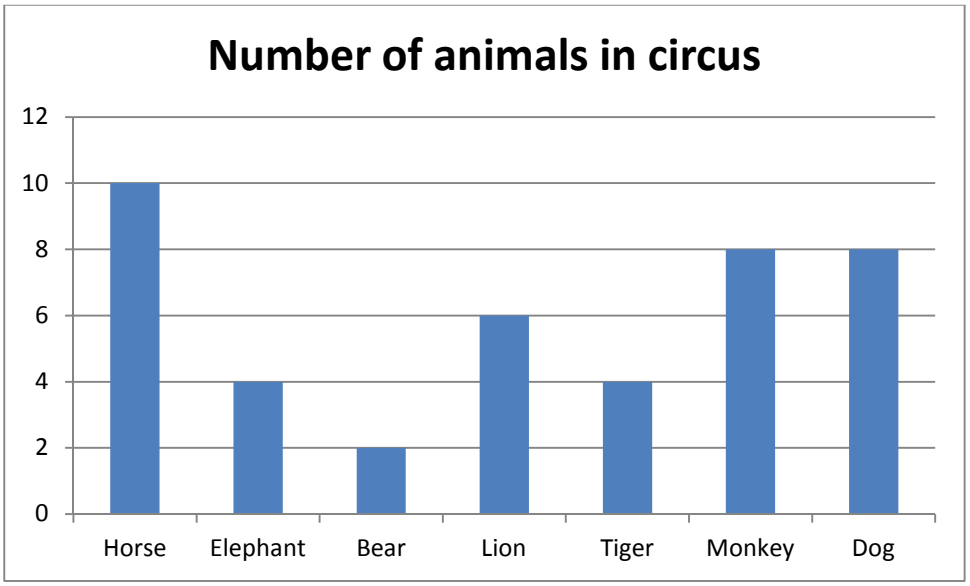
May & September; or March & November

e) Explain why the price changed so much during the year?

More people want to buy blankets in winter and less in summer, so shops put their prices up or down

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4) The graph below shows the number of different animals in a circus



a) What animal is there most of?

Horses

b) What animal is there least of?

Bears

c) How many tigers plus bears are there?

2 + 4 = 6

d) How many different types of animals are there?

7

e) What animal are there exactly 6 of?

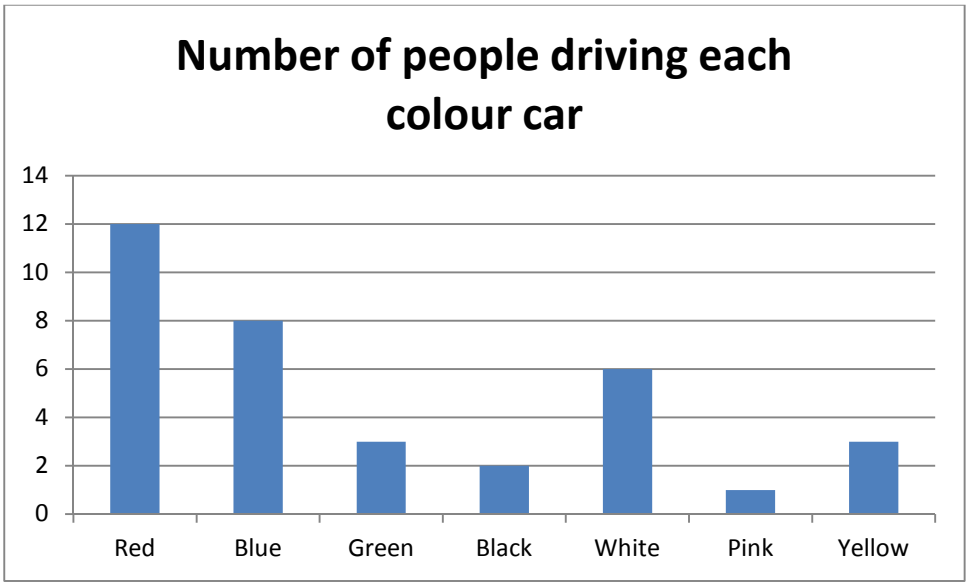
Lions

f) How many animals in total in the circus?

10 + 4 + 2 + 6 + 4 + 8 + 8 = 42

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5) The graph shows the number of people that own a certain colour car



a) How many people drive a green car?

3

b) Which colour car do the least number of people drive?

Pink

c) How many people were asked the question?

12 + 8 + 3 + 2 + 6 + 1 + 3 = 35

d) Which colour is the second favourite?

Blue

e) What colour car do exactly 2 of the people drive?

Black

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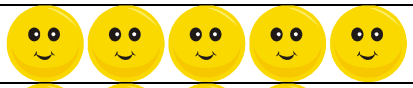
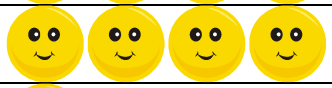
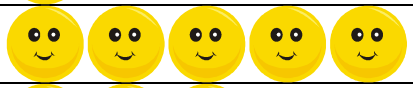
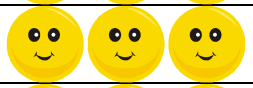
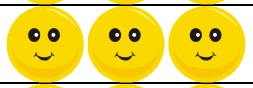
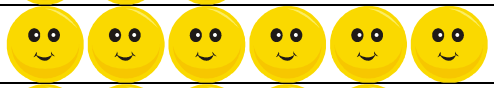
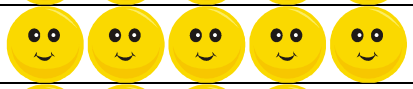
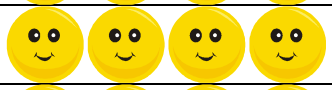
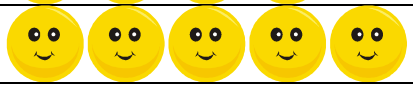
Exercise 2

Picture Graphs

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1) The picture graph below shows a sport and the number of children for whom it is their favourite

Each "face" represents 2 people

Game Number	Attendance
Football	
Rugby	
Soccer	
Basketball	
Hockey	
Swimming	
Tennis	
Golf	
Bowling	
Baseball	

a) Which sport is most popular?

Tennis

b) For how many people is it their favourite?

6 x 2 = 12

c) For how many people is swimming their favourite sport?

3 x 2 = 6

d) How many people were asked?

41 x 2 = 84

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e) Is swimming or hockey more popular?

They are equally popular

2) Some people were asked how many times they ate fish. The picture graph shows their answers. Each fish represents 10 days of the year

Name	Number of days eating fish
Tom	
Benny	
Jane	
Julie	
Karen	
Brian	
Richard	
Ray	
Daniel	
Craig	

a) Who eats fish the most days of the year?

Jane

b) How many days a year do they eat fish?

8 x 10 = 80

c) Who eats fish on the least number of days?

Richard

d) How many days do they eat fish on?

2 x 10 = 20

e) If someone ate fish on 45 days of the year, how could you show this on the graph? Can you think of a better way to show numbers of days that are not groups of 10?

Half a fish could show 5 days, so 45 could be shown with and a half fish

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3) A student went to all the shops in his area and found which of the 5 fruits below was most expensive in that shop. Each piece of fruit is a shop where that fruit was most expensive



a) In how many shops were strawberries the most expensive?

4

b) What fruit was the most expensive in only 2 shops?

Oranges

c) How many shops did the student visit?

17

d) What fruit was most expensive in more shops?

Bananas

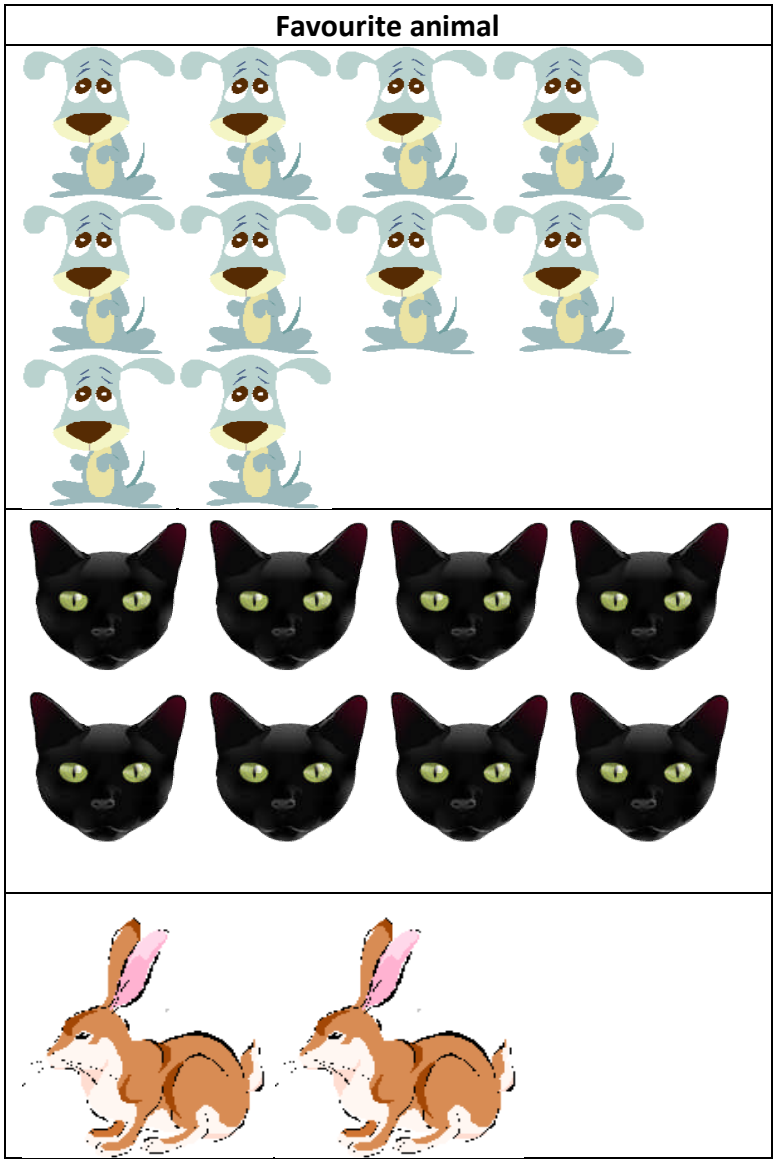
e) What could the student have done if he found a new shop in which pears were the most expensive?

Add another column to his graph for pears

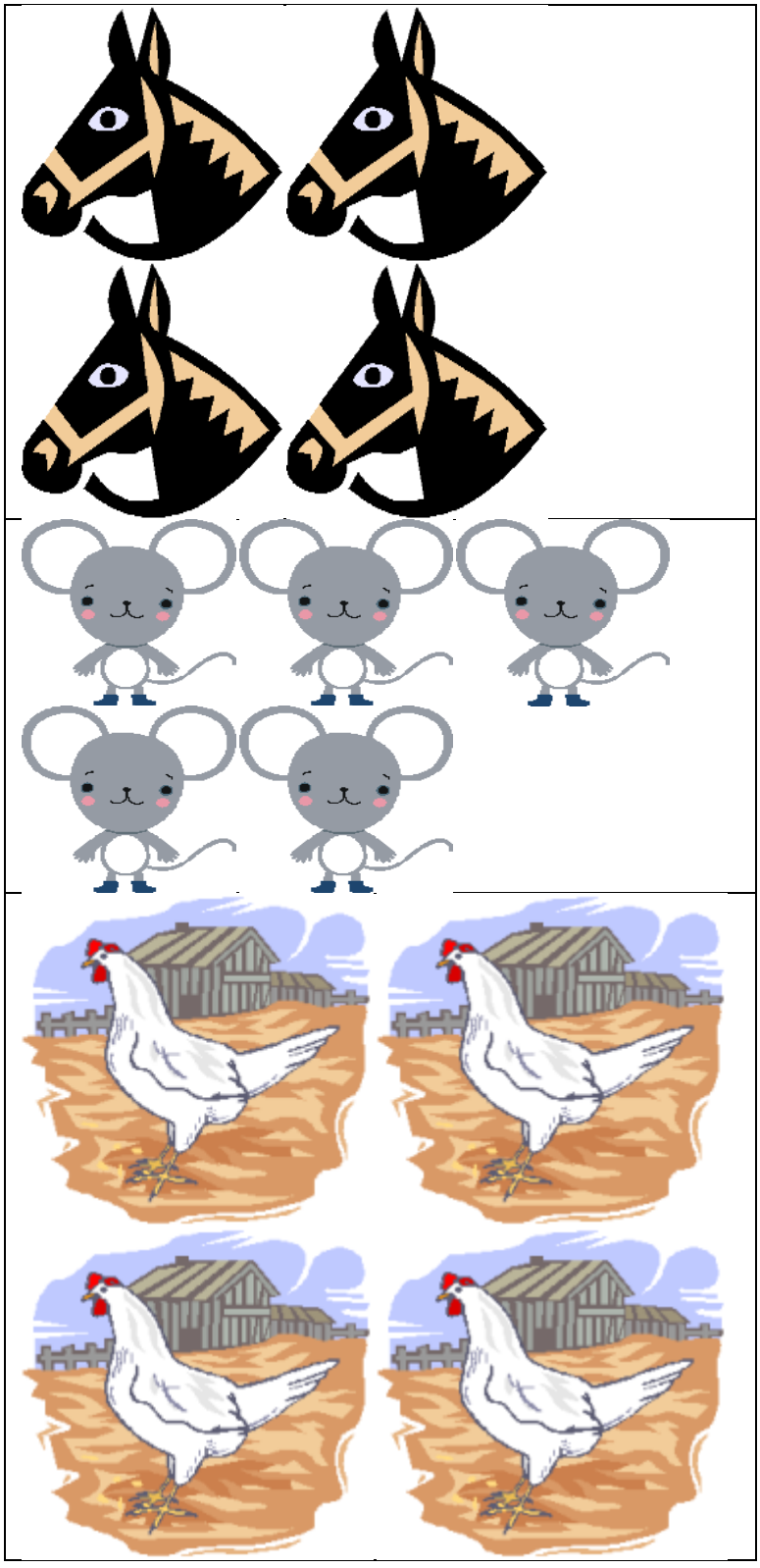
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4) Draw a picture graph that shows the number of people that voted for their favourite animal

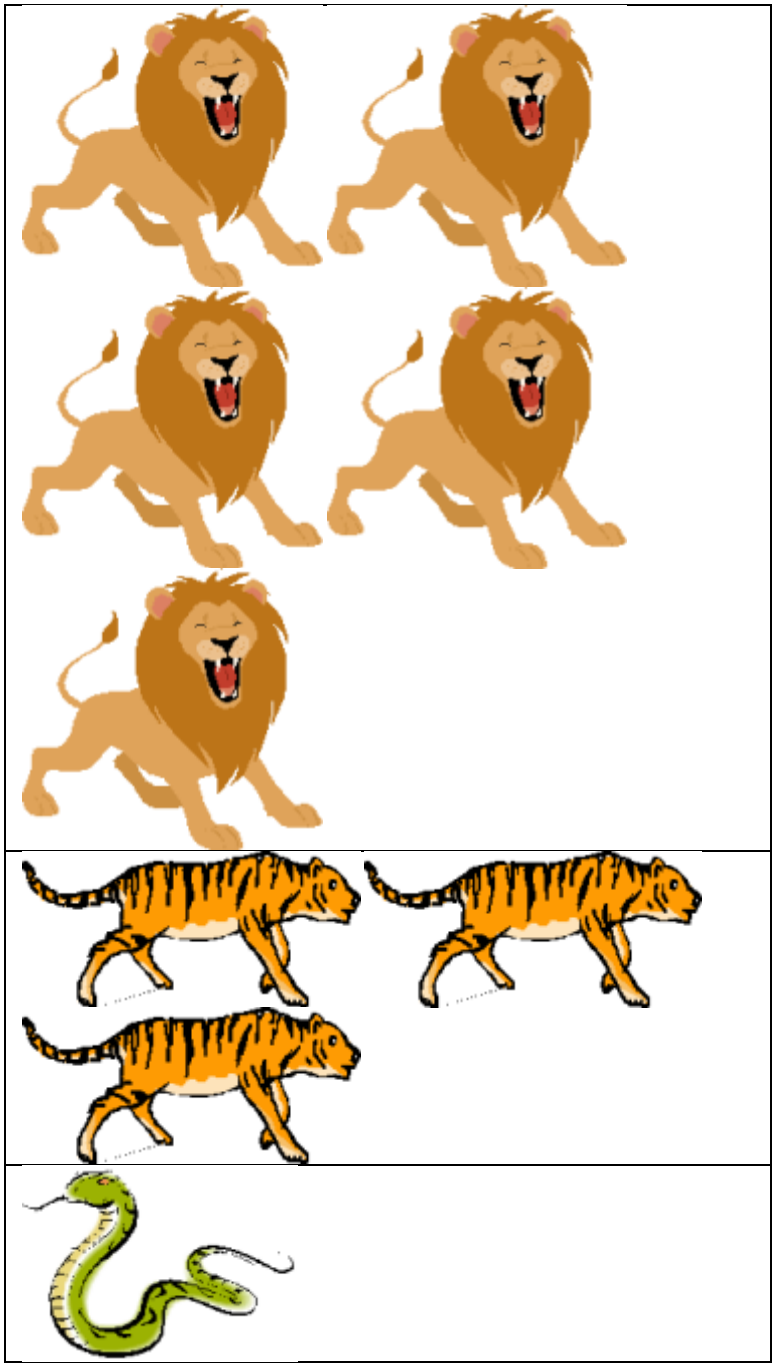
Animal	Number of people
Dog	10
Cat	8
Rabbit	2
Horse	4
Mouse	5
Chicken	4
Lion	5
Tiger	3
Snake	1
Monkey	0



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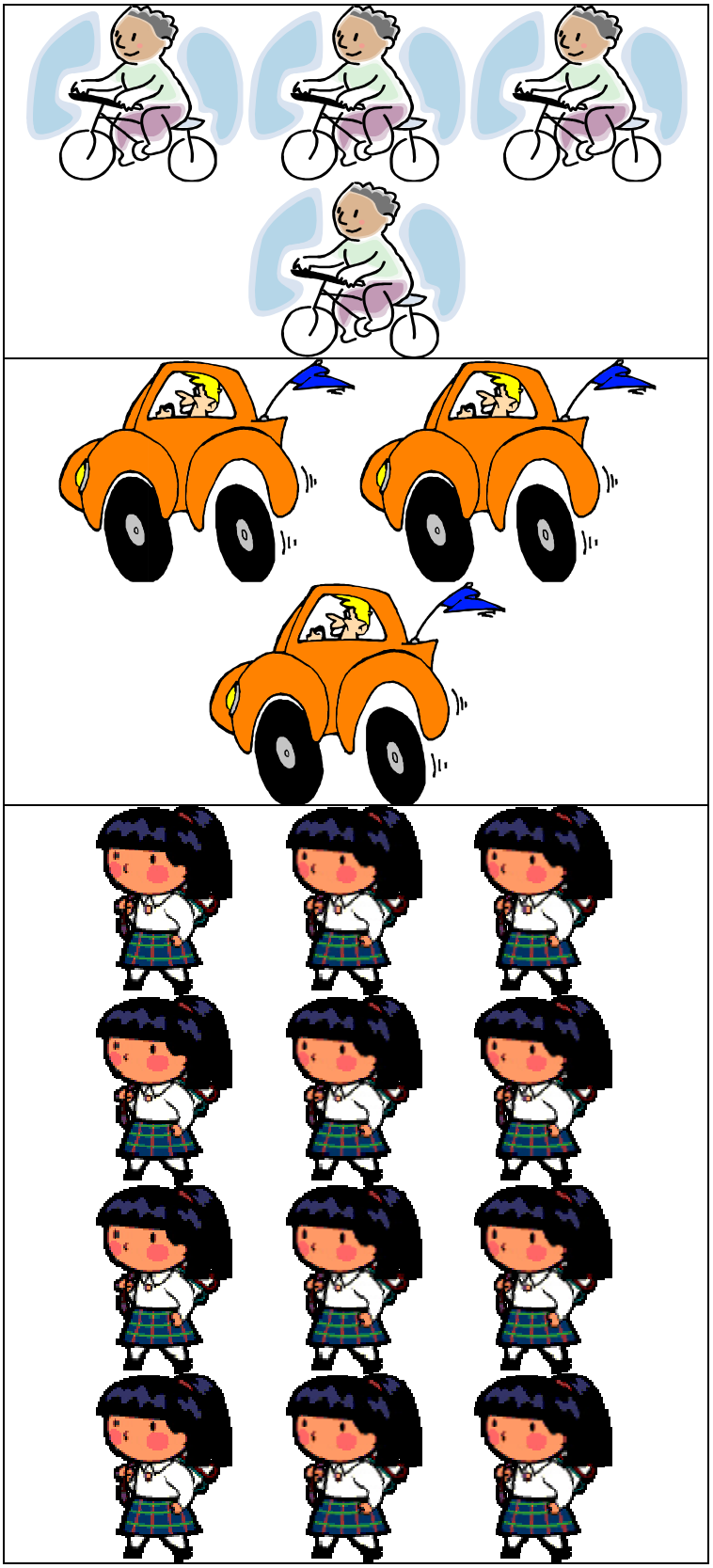
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5) Describe what this picture graph might be showing



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The number of children that get to school in different ways

Bus: 5

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Ride: 4
Driven: 3
Walk: 12

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Year 3 Mathematics

Solutions

Shapes

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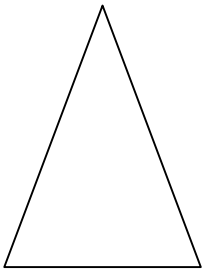
Exercise 1

Common 2D Shapes

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1) Name these shapes

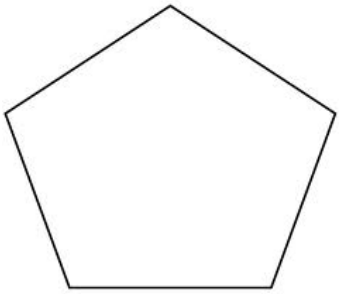
a) Triangle



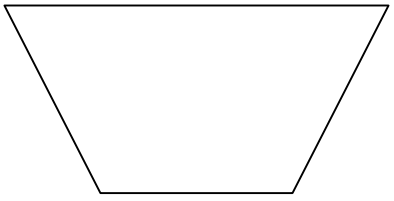
b) Square



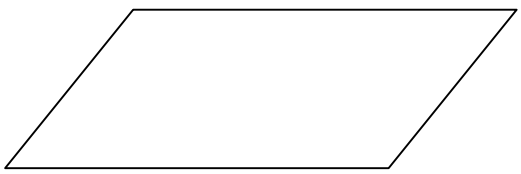
c) Pentagon



d) Trapezium



e) Parallelogram

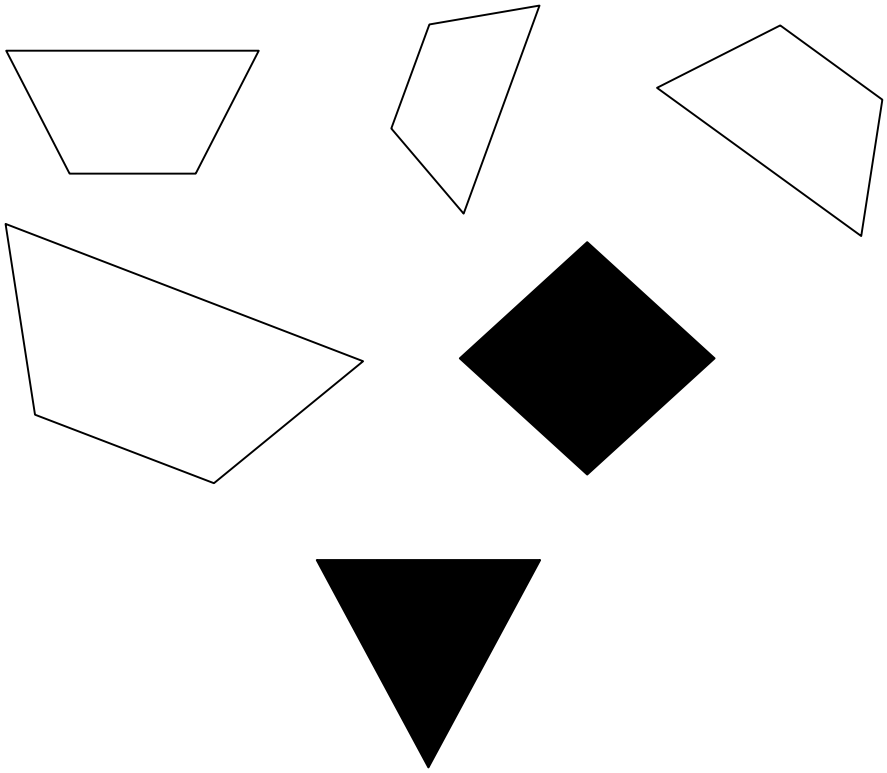


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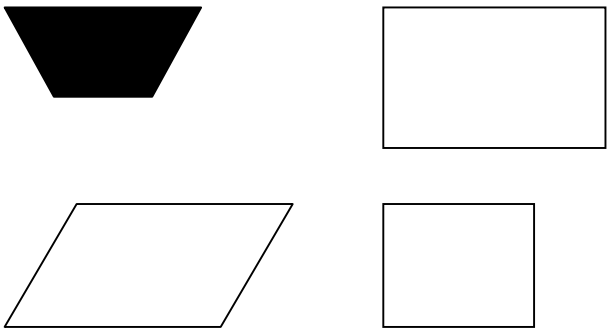
2) In the space in the table, write down how many sides each of the shapes has

Triangle	3
Square	4
Rectangle	4
Parallelogram	4
Pentagon	5
Trapezium	4

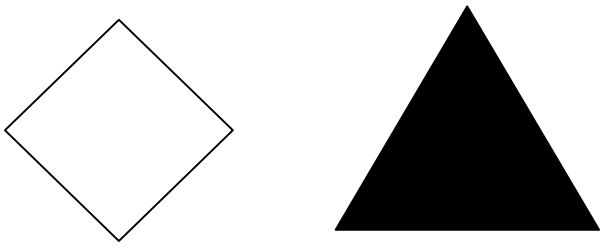
3) Which of the following shapes is NOT a trapezium?



4) Which of these shapes is NOT a parallelogram?



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5)

a) What do a square, rectangle, parallelogram and a trapezium all have in common?

They each have 4 sides

b) What makes a trapezium different to the other three?

Two sides of a trapezium are not parallel

c) What makes a rectangle different to a parallelogram?

All the angles of a rectangle are the same

d) What makes a square different to a rectangle?

All sides of a square are the same length

e) Do you think that a square is a special type of rectangle, or that a rectangle is a special type of square?

A square is a special type of rectangle that has all sides equal. A rectangle cannot be a square, since a square must have equal sides

6)

Name the shape from the descriptions (could be more than one name)

a) It has 4 sides

Square, rectangle, parallelogram, trapezium

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b) It has 4 sides with both pairs of sides parallel

Square, rectangle, parallelogram

c) It has 4 sides with both pairs of sides parallel and all corners are right angles

Square, rectangle

d) It has 4 sides with both pairs of sides parallel and all corners are not right angles

Parallelogram

e) It has 4 sides with both pairs of sides parallel, all sides the same length and all corners are right angles

Square

f) It has 4 sides with one pair of sides parallel

Trapezium

g) It has 5 sides

Pentagon

7) Name the shapes that you can see in this picture



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Trapezium (chimney)
Triangle (roof)
Rectangle: Door, Wall
Square: Window, window pieces
How many squares are in this picture?

5: the window and each of the pieces

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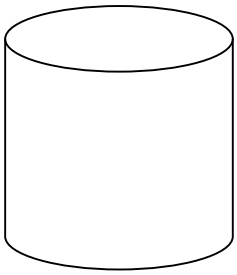
Exercise 2

Simple 3D Shapes

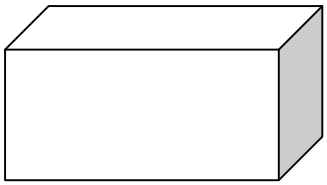
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1) Name these shapes

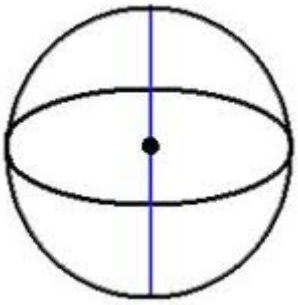
a) Cylinder



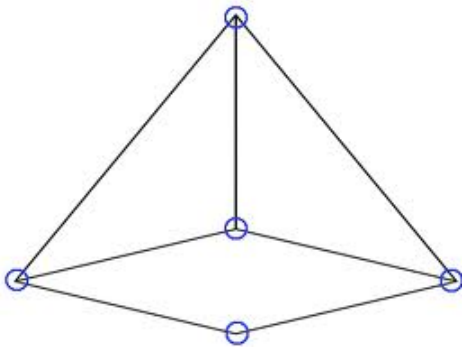
b) Rectangular prism



c) Sphere

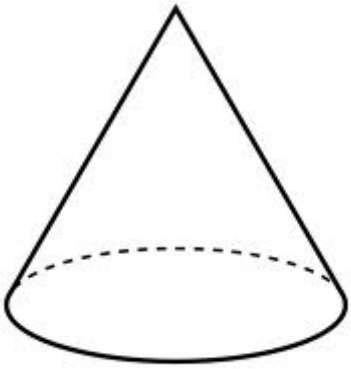


d) Square base pyramid



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e) Cone



2) What is the main difference between a cone and a pyramid?

A cone has a circular base, a pyramid does not

3) Sort the following shapes into two groups; those that can have a square base and those that cannot

Pyramid, Sphere, Cone, Cylinder, Prism

Square base possible: Pyramid, prism
Cannot have square base: Sphere, cone, cylinder

4) Which of the above shapes can have a point at the top (an apex)?

Pyramid, cone

5) Which of the above shapes best describes the following?

a) The shape of the Earth?

Sphere

b) The shape of an ice cream?

Cone

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c) The shape of a can of soup?

Cylinder

d) The shape of a cardboard box?

Prism

6) What is the same about a cone and a cylinder?

They have circular bases

7) If you had a cylinder and a cone that had the same size base and were the same height which one could you fit more water into?

The cylinder, since the cone gets smaller as it gets taller

8) Name the shape that these everyday objects are made from

a)



Sphere

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b)



Cone

c)



Cylinder

d)



Pyramid

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Year 3 Mathematics

Measurement

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Exercise 1

Time

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1) What time is it on the following clocks?

a)



2 o'clock

b)



3 o'clock

c)



8 o'clock

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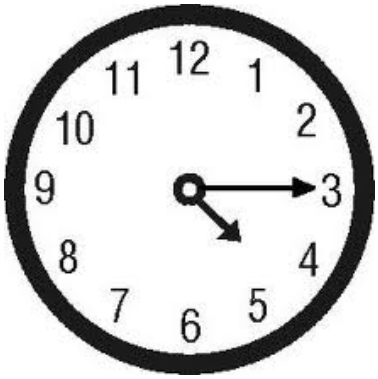
d)



11 o'clock

2) What time is it on the following clocks?

a)



Quarter past 4

b)



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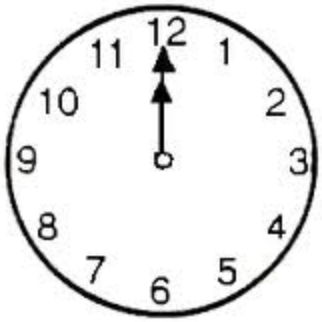
Half past 8

c)



Quarter to 8

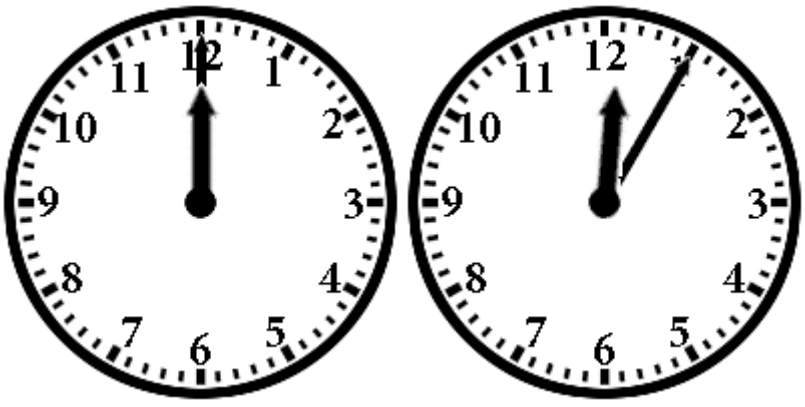
d)



12 o'clock

3) How much time has gone by between each of the two clocks?

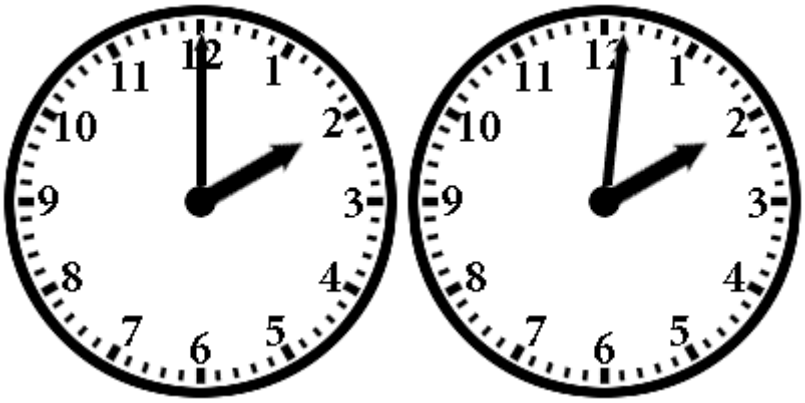
a)



5 minutes

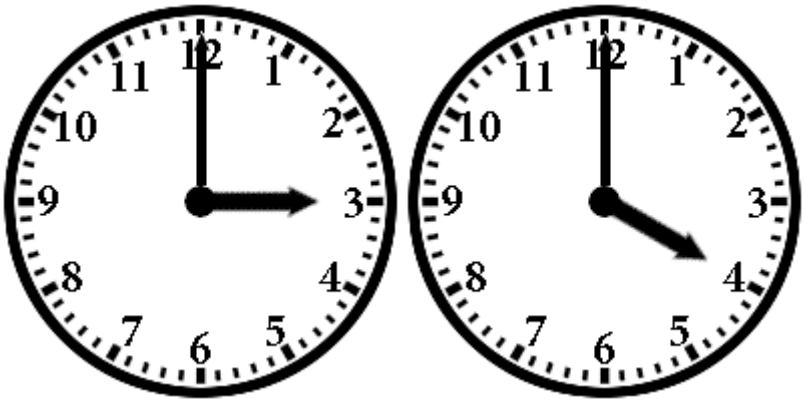
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b)



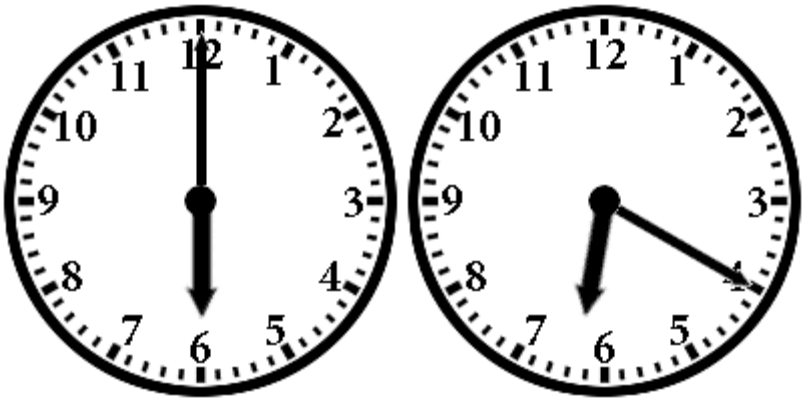
One minute

c)



One hour

d)



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20 minutes

4) Draw the following times on a clock

a) One fifteen



b) Quarter to three



c) Half past two



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d) Two thirty



e) Quarter past four



5) Peter started homework at 6 o'clock and had it all finished by half past 6. How much time had he spent on his homework?

30 minutes

6) Karen went to bed at eight fifteen, and Robert at seven forty five. How many minutes were there between their bedtimes?

30 minutes

7) John's dad leaves for work at seven o'clock in the morning and gets to work at half past seven. How long does it take John's dad to get to work?

30 minutes

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8) Bill's mum was exercising. She started at five fifteen and had to exercise for 45 minutes. At what time should she stop exercising?

6 o'clock

9) A worker gets fifteen minutes for morning tea. If he starts morning tea at 8:45, what time should he start work again?

9 o'clock

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Exercise 2

Mass

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1) Which of the following would weigh about 1kg?

- One litre of water

  Yes

- An elephant

  No

- A car

  No

- A baseball bat

  Possibly

- A fly

  No

- Two tubs of margarine

  Yes

- A person

  No

- A biscuit

  No

- Bag of sugar

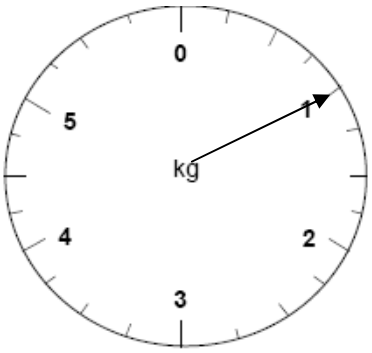
  Yes

2) Tom has 3 objects that each weighs 1 kg. How much do the objects weigh in total?

3 kg

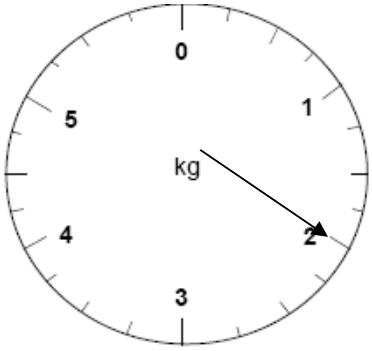
3) Draw a pointer on the scale to show the following masses

a) 1 kg

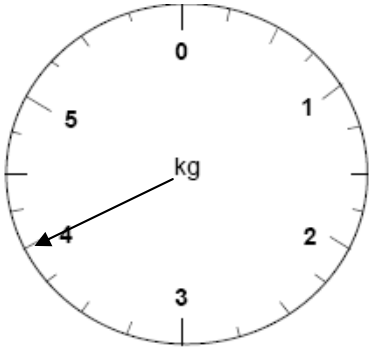


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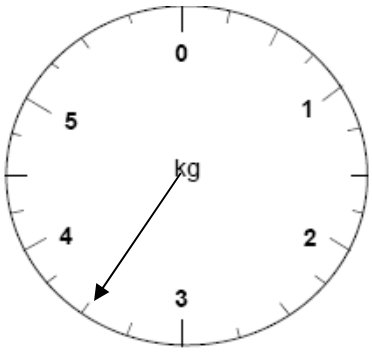
b) 2 kg



c) 4 kg

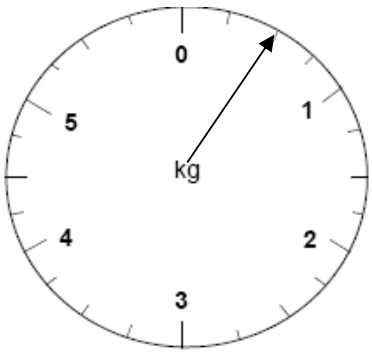


d) 3 and a half kg



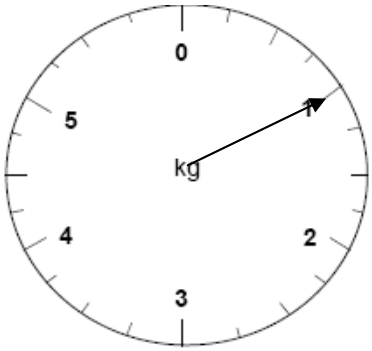
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e) Half a kg



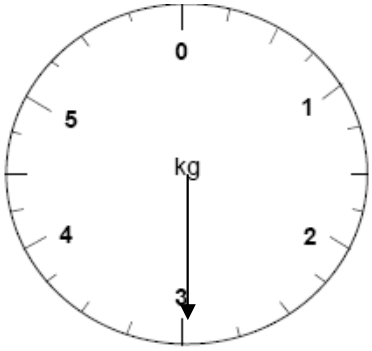
4) Write down the reading on each scale

a)



1 kg

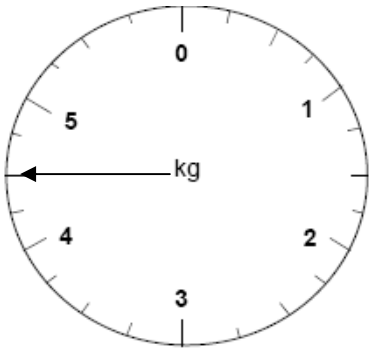
b)



3 kg

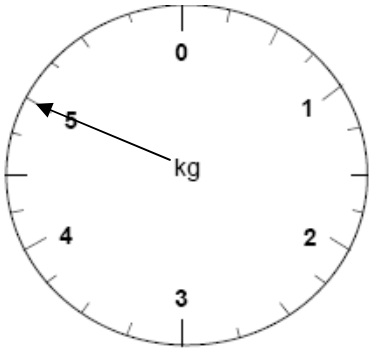
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c)



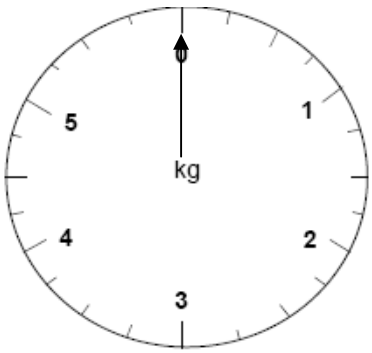
4 and a half kg

d)



5 kg

e)



Zero (6 kg is an acceptable answer)

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5) Eric has some margarine tubs that each has a mass of ½ kg. How many would he need to place on a scale so it balances with objects that have a mass of:

a) 2 kg

4

b) 1 and ½ kg

3

c) 4 kg

8

d) 3 kg

6

6) What has more mass:

a) A truck or a bicycle?

Bicycle

b) A man or a giraffe?

Giraffe

c) A fly or a plane?

Plane

d) A mobile phone or a computer?

Computer

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e) A packet of biscuits or a carton of soft drink?

A packet of biscuits

f) A piece of string or a jumper?

Jumper

7) Mr Jones got two of his students to help move sand in a wheelbarrow. Mr Jones moved 25 kg, Robert moved 10 kg, and Alex moved 8 kg. How much sand did they move altogether?

25 + 10 + 8 = 43 kg

8) Sausages cost $2 for 3 kg; how much would 6 kg of sausages cost?

There are two lots of 3 kg, so the cost would be 2 x $2 = $4

9) Eric's father weighs 3 times as much as Eric. If his father weighs 75 kg, how much does Eric weigh?

3 x Eric's weight = 75 kg

Therefore Eric weighs 25 kg

10) John has one 1 kg weight, three 2 kg weights, and one 5 kg weight What would he need to place on a scale to balance:

a) 5 kg

Either the 5 kg weight OR two of the 2 kg and the 1 kg weights

b) 6 kg

Either the 5 and 1 kg weights OR the three 2 kg weights

c) 7 kg

Either the 5 kg and one of the 2 kg weights OR the three 2kg and 1 kg weights

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d) 9 kg

The 5 kg and two of the 2 kg weights

e) 12 kg

All of the weights

f) 15 kg

He does not have enough weights to balance 15 kg

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Exercise 3

Length

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1) How many centimetres in:

a) 1 meter

100

b) ½ metre

50

c) 2 metres

200

d) 3 ½ metres

350

e) 5 metres

500

2) How many metres in:

a) 50 cm

½

b) 100 cm

1

c) 200 cm

2

d) 250 cm

2 ½

e) 550 cm

5 ½

3) Graham is 145 cm tall, while his dad is 2 metres. How much taller is Graham's dad?

2m = 200cm

200cm – 145cm = 55cm

4) How many 1 meter rulers would need to be laid end to end to measure a length of 375 cm?

100cm = 1m

375cm = 3.75m

So need 4 rulers

5) Mark rolled a ball 5m 40cm, while his friend rolled a ball 6m 10 cm. How much further did the second ball travel?

5m 40cm = 540cm

6m 10cm =610cm

610cm – 540cm = 70cm

6) Geoffrey is painting a 4 metre line. A full paintbrush paints 20 cm before it has to be dipped back into the can. How many times does Geoffrey have to dip his brush in the can to paint the line?

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4m =400cm

400cm ÷ 20cm = 20

7) Two pieces of wood are joined together length ways. The first piece is 2 m 60 cm long; the second piece is 3 m 60cm long. How long is the new piece of wood?

2m 60cm = 260cm

3m 60cm = 360cm

260cm + 360cm = 620cm
= 6m 20cm

8) James took a pace and measured it to be 80 cm, while his dad's pace is 1m 30 cm; how far would they step one after the other?

1m 30cm = 130cm

130cm + 80cm = 210cm

210cm = 2m 10cm

9) Half the length of a piece of rope is 2m 75 cm. How long is the piece of rope?

2m 75cm + 2m 75cm = 4m 150cm

= 5m 50cm

10) A horse is 2m 70 cm tall, while its rider is 1m 50cm. How far off the ground is the rider when he is sitting on the horse?

2m 70cm = 270cm

1m 50cm = 150cm

270cm + 150cm = 420cm

420cm = 4m 20cm

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Exercise 4

Area

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1) The picture shows a block that is 1 cm long and 1 cm wide



a) What is the common name for this sized square?

1 cm^2

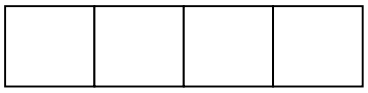
b) Estimate how many of these squares it would take to fill

I. Your computer screen

II. Your exercise book
III. Your desk
IV. Your bedroom floor

2) How many cm^3 is shown in each of the following diagrams?

a)



4

b)



8

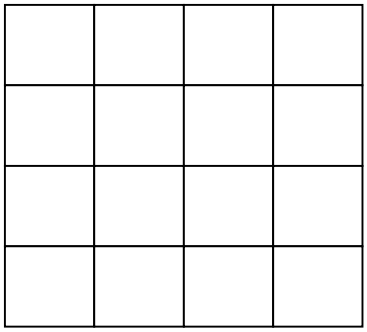
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c)



4

d)



16

e) Explain why the answers to parts a and c are the same although the total shape is different

Different shapes can have the same area

3) Make, draw or imagine a grid with 10 rows and 10 columns of 1 cm squares. Which of the following would your grid be too small to fit, which would your grid be too big for, and which would it fit pretty close to exactly?

a) A school book

Possibly fit

b) A floor

Too small

c) A back yard

Too small

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d) The lid of a laptop

Possibly fit

e) A calculator

Too big

f) A stamp

Too big

g) A chessboard

Probably fit

4) Put the following in order from smallest to largest area

- The roof of a car
- A plate
- An IPod
- The side of a house
- A coin
- Australia

Coin

IPod

Plate

Car roof

Side of a house

Australia

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5) Peter made a grid of 1 cm squares. His grid was 5 rows by 4 columns. John made a grid that was 3 rows by 6 columns.

a) Whose grid was bigger and by how many squares?

Peter by 2 squares

b) Graham made a grid that was 10 rows and 2 columns. Whose grid was the same area as Graham's?

Peter's and Graham's would both have 20 squares

c) Graham's grid fitted exactly over a book; why wouldn't either of Peter's or John's grids fit exactly over the same book, even though one of them is the same area?

John's is too small, while Peter's is a different length and width

d) How many of Peter's grids would be needed to cover an object that was 20 rows long and 16 columns wide?

20 x 16 = 5 lots of 5 x 4

Would need 5 of Peter's grids

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Exercise 5

Volume

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1) Which of the following could hold 1 litre of liquid or more?

- **A milk carton**
- **A car's petrol tank**
- A teacup
- **A bath**
- A straw
- **A swimming pool**
- A teaspoon

2) Put the following in order from smallest to largest capacity

- A tablespoon
- A spa bath
- A dog's drinking bowl
- A shampoo bottle
- An ocean
- An eye dropper

Eye dropper

Tablespoon

Shampoo bottle

Dog's bowl

Spa bath

Ocean

3) Each of the small cubes measures 1cm x 1cm x 1cm.

a) What is the common name for such a cube?

1 cm^3

b) Why is a cube useful for measuring volumes of objects?

They stack neatly and can be counted quickly

c) Name an object that would take about 12 cubes inside

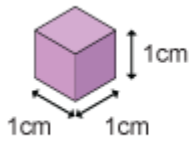
d) Name an object that would be too big to measure the volume of by just using these cubes

Example answer: A house

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4) Each cube is 1cm x 1cm x 1cm. What is the volume of each of the stacks?

a)



1 cm^3

b)



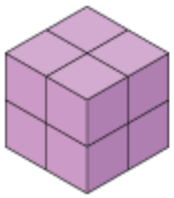
3 cm^3

c)



4 cm^3

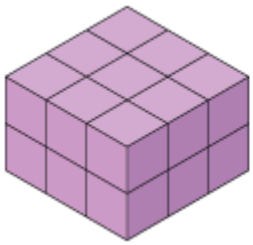
d)



8 cm^3

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e)



18 cm^3

f) Explain why the answer to part c and d can be the same even though the stacks look different

Different shapes can have the same volume

5) Stacks of 1 cm blocks are built. How many blocks are in each of the following stacks?

a) 2 rows and 3 columns

6

b) 4 rows and 5 columns

20

c) 6 rows and 3 columns

18

d) 3 rows and 6 columns

18

e) 10 rows and 10 columns

100

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f) 30 rows and 30 columns

900

g) Which of the above stacks do you think would nearly fill a one litre container?

30 rows and 30 columns

6) Mark had a stack of 1 cm blocks that was 5 rows and 6 columns. Peter's stack was twice as long and twice as wide. How many more blocks did Peter's stack have in it than Mark's?

Peter's stack was 10 rows and 12 columns = 120 blocks

Mark's stack was 5 x 6 = 30 blocks

Peter had 90 more blocks

7) Alan had a stack of 1 cm blocks that was 10 rows and 4 columns. David's stack had half as many rows and half as many columns. How many less blocks did David's stack have?

Alan had 10 x 4 = 40 blocks

David had 5 rows and 2 columns = 10 blocks

David had 30 less blocks

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Year 3 Mathematics

Solutions

Space

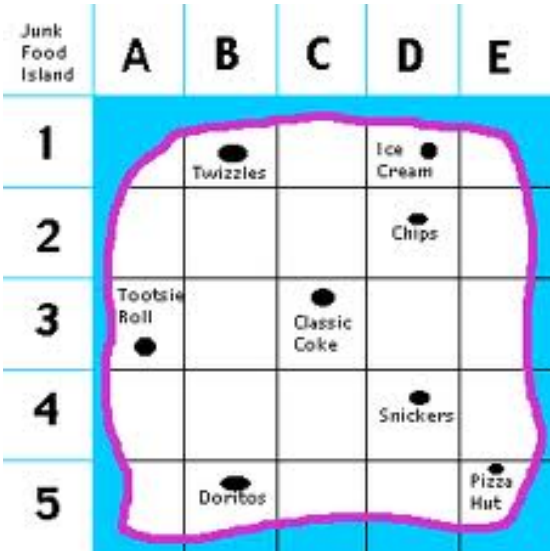
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Exercise 1

Map Legends & Directions

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1) Answer the following questions about Junk Food Island



a) Where would you find chips?

D2

b) What is located at B1

Twizzles

c) Name a grid location where there is no junk food

Example answer: B2

d) Name a grid location where there is mostly water

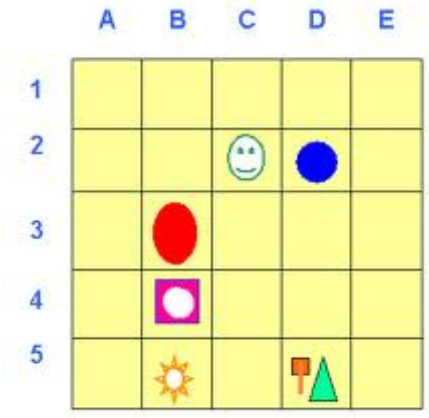
Example answer: A1

e) If you started at Doritos and walked right until you came to the next grid location of junk food, what grid location would you be at?

E5

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2)



a) At what grid location would you find a smiley face?

C2

b) At what grid location would you find a triangle

D5

c) What is at grid location B5?

Sun

d) Which row contains no symbols?

Row 1

3) The map shows some streets in Sydney



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a) Central Station is at the end of which street?

Rawson

b) Which street joins Pitt St, George St and Sussex St?

Hay

c) "We are here" on the corner of which two streets?

Ultimo and George

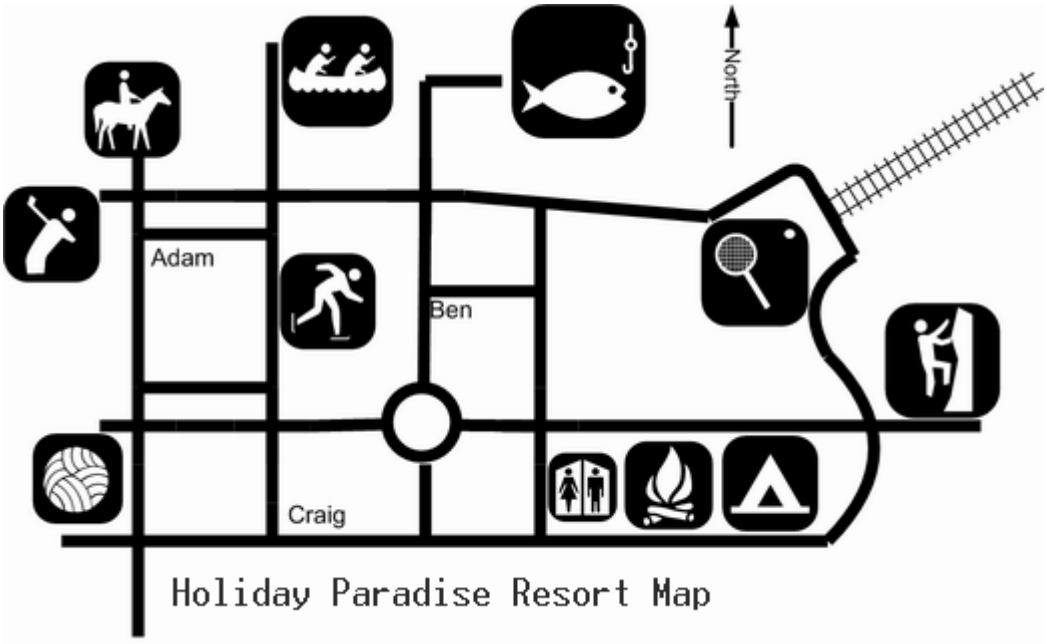
d) If you go left from "We are here" which street do you go down?

Ultimo

e) Which street is Chinatown near?

Sussex

4) Use the map of Holiday Paradise Resort to answer the questions



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a) Whose house is closest to the golf?

Adam

b) If you started at Craig's house and walked north as far as you could go, what would you find?

Canoeing

c) What sport is played near the railway line?

Tennis

d) Describe how you would get from horse riding to fishing (the quickest way)

Head south
Take first left
Take second left
Take first right

e) Start at Ben's house facing north, walk north and turn left at the next street. Turn left at the next street and go all the way to the end. Where do you finish?

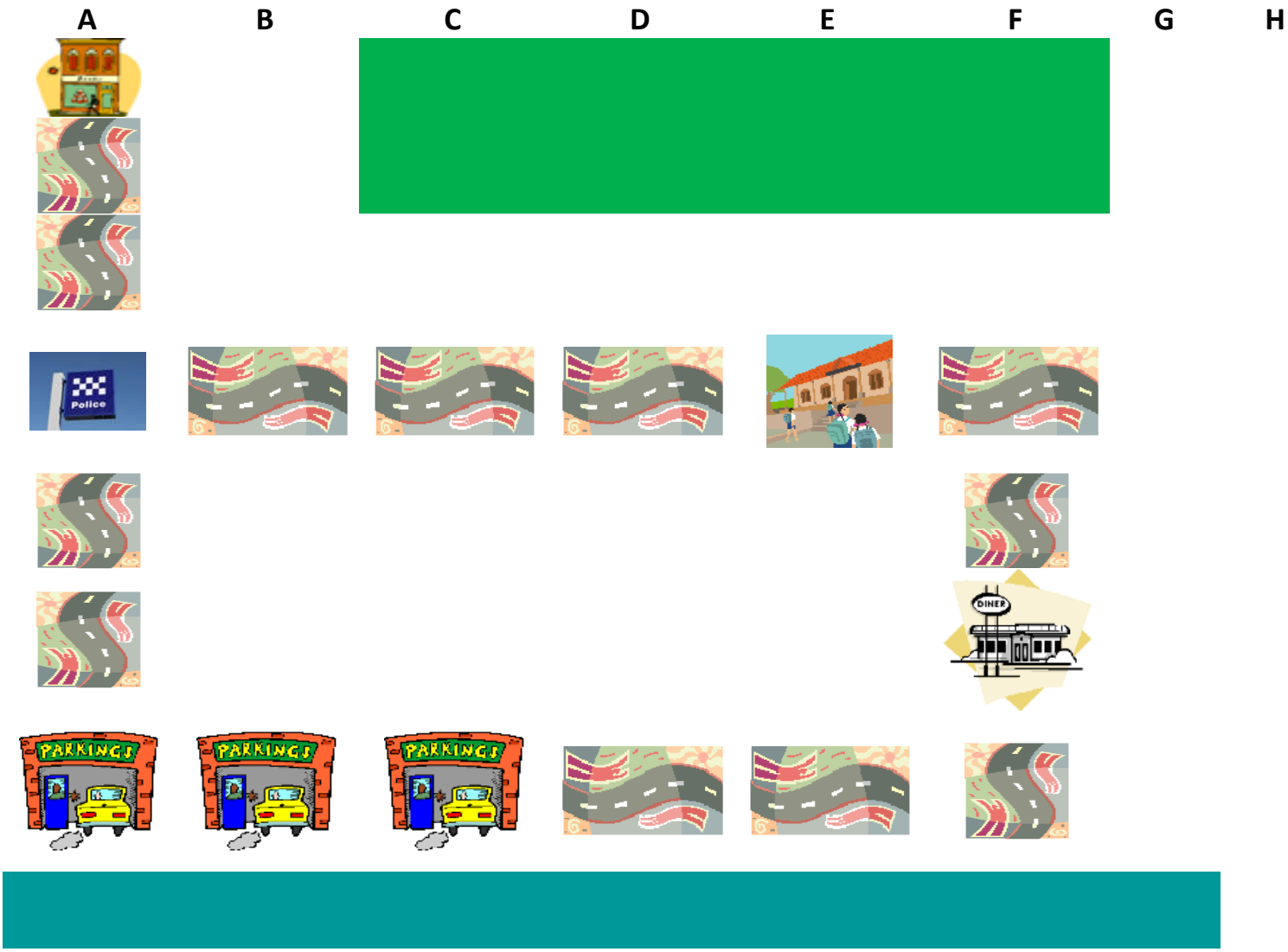
Craig's house

f) Even though it is not shown on the map, where do you think the lake is, and why?

Top middle of the map; this is where the canoeing and fishing are

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5) Record the following information on the grid



a) There is a shop at A1

b) Three squares down from the shop is the police station

c) From the police station go toward the right of the page 4 squares and draw a school

d) A river runs from A8 to G8

e) There is a car park running from A7 to C7

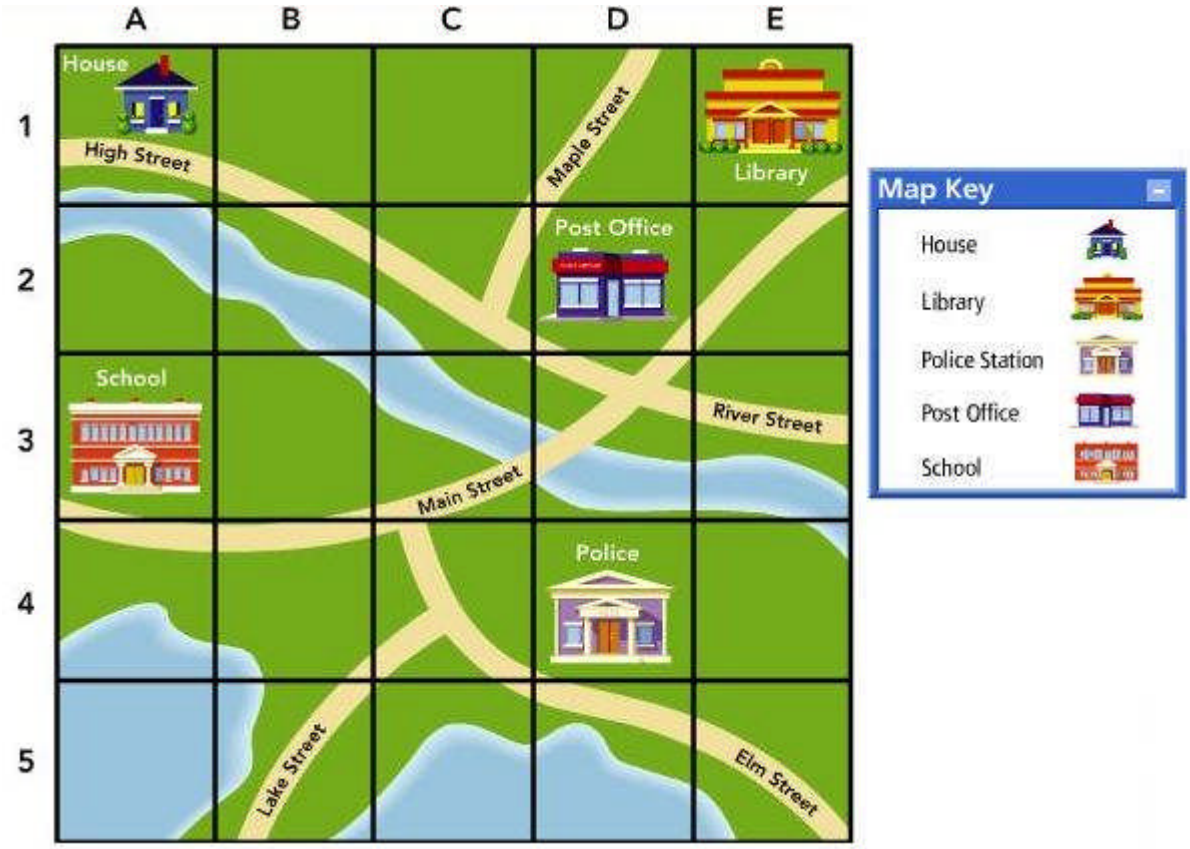
f) At H6 there is a restaurant

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g) Draw roads that connect all the major parts of the map

h) On any part of the map separated from buildings or roads by one square or more draw parkland

6)



a) Where is the post office located?

D2

b) River Street changes its name to what street?

High Street

c) What building is at D4?

Police Station

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d) Name all the grid points that have at least part of the river in them

A1, A2, B2, B3, C2, C3, D3, E3, E4

e) At what point does Lake Street meet Elm Street?

C4

f) To get from the school to the library, what street would you need to take?

Main Street

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